

Regularized Multivariate Two-way Functional Principal Component Analysis

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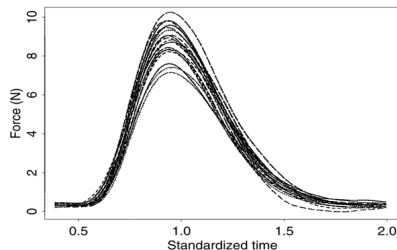
Outline

- 1 Introduction & Background
- 2 A SVD Approach for Regularized Multivariate FPCA
- 3 Two-way Regularized Multivariate FPCA
- 4 Conclusion & Future Work

Introduction

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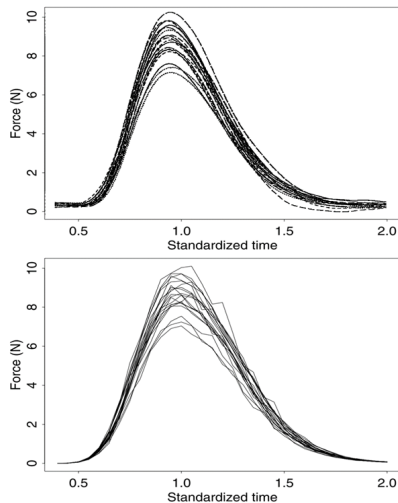
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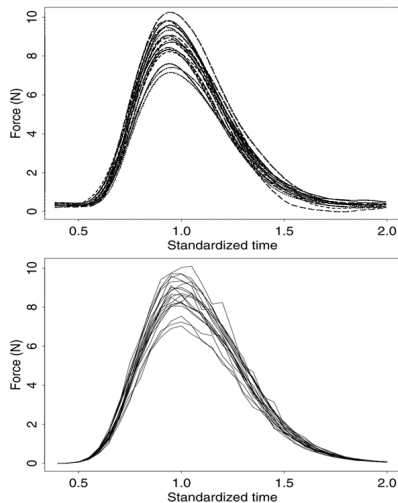
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- These data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.
- Functional Data Analysis (FDA) is a statistical framework that treats these observations as realizations of smooth underlying functions, allowing for more accurate modeling and interpretation of continuous processes.



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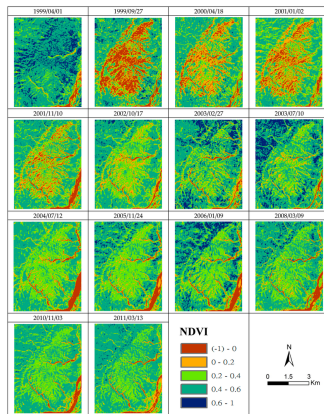
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- Impact: More adaptable, robust, and applicable across diverse scientific and business problems.

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 - Effective for dimension reduction in complex datasets.

Mathematical Foundation of FPCA

- **Goal:** Identify functional directions that maximize variance (low-rank approximation of functional data). For functional data $X \in \mathbb{R}^{n \times m}$ contains the discretized functional observations (rows correspond to subjects, columns to grid points), $v \in \mathbb{R}^m$ represents the estimated principal component (function), and $u \in \mathbb{R}^n$ denotes the associated principal component scores.

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- **Optimization steps:**

$$\text{Fix } v : u = \frac{Xv}{v^\top v} \quad \text{and} \quad \text{Fix } u : v = \frac{X^\top u}{u^\top u}$$

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- Algorithms: Based on iterative **power method** and **thresholding** updates.

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- A power algorithm is defined to compute the PCs while incorporating smoothness penalty.
- Consider the SVD of X as $X = UDV^\top$, where U and V have orthonormal columns and D is diagonal with ordered singular values. In particular, for $X = u_d v_d^\top$, v is the first principal component and $u = u_d$ gives the associated scores. With these representations, the power algorithm (described below) converges quickly, typically in only a few iterations.

Power Algorithm

Algorithm: Penalized Power Iteration

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- For notational convenience, we define $S(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{m \times m}$, which simplifies expressions involving regularization. The penalty matrix Ω is set up so that larger values of the quadratic form $v^\top \Omega v$ mean rougher functions. This means that it penalizes functions that change quickly between time points.

Tuning Smoothness Parameters

- To select the optimal tuning parameters α efficiently, one can use a traditional Cross-Validation (CV) criterion and a computationally efficient closed-form Generalized Cross-Validation (GCV) criterion:

$$\text{CV}(\alpha) = \frac{1}{m} \sum_{j=1}^m \frac{\left[\{(I - S(\alpha))(X^T u)\}_{jj} \right]^2}{(1 - \{S(\alpha)\}_{jj})^2},$$

where $\{\cdot\}_{jj}$ denotes the j -th diagonal element.

$$\text{GCV}(\alpha) = \frac{1}{m} \frac{\|(I - S(\alpha))(X^T u)\|^2}{\left(1 - \frac{1}{m} \text{tr}\{S(\alpha)\}\right)^2}.$$

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- Sparse FPCA formulation [Shen and Huang, 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + p_\gamma(v)$$

where $p_\gamma(v)$ is a sparsity-inducing penalty.

Sparsity penalties

- **Lasso (Soft-thresholding):**

$$p_{\gamma}^{\text{soft}}(|\theta|) = 2\gamma|\theta|, \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{soft}}(y) = \text{sign}(y)(|y| - \gamma)_+$$

- **Hard thresholding:**

$$p_{\gamma}^{\text{hard}}(|\theta|) = \gamma^2 I(|\theta| \neq 0), \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{hard}}(y) = I(|y| > \gamma) y$$

- **SCAD penalty:**

$$p_{\gamma}^{\text{SCAD}}(|\theta|) = \begin{cases} 2\gamma|\theta|, & |\theta| \leq \gamma, \\ \frac{\theta^2 - 2a\gamma|\theta| + \gamma^2}{a-1}, & \gamma < |\theta| \leq a\gamma, \\ \frac{(a+1)\gamma^2}{2}, & |\theta| > a\gamma, \end{cases} \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{SCAD}}(y) = \begin{cases} \text{sign}(y)(|y| - \gamma)_+, & |y| \leq 2\gamma, \\ \frac{(a-1)y - \text{sign}(y)a\gamma}{a-2}, & 2\gamma < |y| \leq a\gamma, \\ y, & |y| > a\gamma, \end{cases}$$

where $a = 3.7$ (Fan and Li [2001]).

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 - ➌ **Normalize Right Singular Vector:** $v \leftarrow \frac{v}{\|v\|}$

Cross-Validation for Sparsity Selection

- **Sparsity parameter:** Tuning parameter controlling number of non-zero loadings in v ($0 =$ dense, $p =$ full sparsity).

Algorithm: K-fold CV Tuning Parameter Selection - Degree of sparsity

- 1 Randomly group the rows of side-by-side data matrix X into K roughly equal-sized groups, denoted as $X^1, \dots, X^K$.

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 - 2 Calculate the K-fold CV scores defined as: (N is the number of grid points in X^k)

$$CV_j = \sum_{k=1}^K \frac{\|X^k - u^{-k}(j)v^k(j)\|^2}{N}$$

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- 3 Select the degree of sparsity as $j_0 = \arg \min \{CV(j)\}$.

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- More stable & applicable to high-dimensional multivariate FDA

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- **Block structure example:**

$$X_i = \begin{bmatrix} x_{1i}(t_{i1}) & \cdots & x_{1i}(t_{i,m_i}) \\ \vdots & \ddots & \vdots \\ x_{ni}(t_{i1}) & \cdots & x_{ni}(t_{i,m_i}) \end{bmatrix}, \quad \mathbf{X} = [X_1 \ X_2 \ \cdots \ X_p]$$

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- The block-diagonal penalty matrix is $\mathbf{\Omega} = \text{blockdiag}(\Omega_1, \Omega_2, \dots, \Omega_p)$, where each $\Omega_i \in \mathbb{R}^{m_i \times m_i}$ is a univariate roughness penalty matrix.

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- The penalized reconstruction error is

$$\min_{u,v} \|\mathbf{X} - uv^\top\|_F^2 + \alpha^\top (v^\top \mathbf{\Omega} v),$$

where $\alpha = (\alpha_1, \dots, \alpha_p)^\top$ controls smoothness.

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- The smoothing operator is $\mathbf{S}(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{M \times M}$.
 - The smoothing parameter α is selected via generalized cross-validation (GCV), defined as

$$\text{GCV}(\alpha) = \frac{1}{M} \frac{\|(I - \mathbf{S}(\alpha))(\mathbf{X}^\top u)\|^2}{\left(1 - \frac{1}{M}\text{tr}\{\mathbf{S}(\alpha)\}\right)^2}. \quad (1)$$

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- Algorithm CV Tuning for Sparsity and equation (1) are used to tune the sparsity level via K-fold CV and the smoothing parameter via GCV, respectively.

Simulation: Estimation Performance

- **Data-generating process:** Two functional variables:

$$X_{ij}^{(1)} = u_{i1}v_{11}(t_j) + u_{i2}v_{12}(t_j) + \epsilon_{ij}^{(1)}, \quad X_{ij}^{(2)} = u_{i1}v_{21}(t_j) + u_{i2}v_{22}(t_j) + \epsilon_{ij}^{(2)},$$

- where $u_{i1} \sim N(0, \sigma_1^2)$, $u_{i2} \sim N(0, \sigma_2^2)$, $\epsilon_{ij}^{(k)} \sim N(0, \sigma^2)$, and $n = m = 101$, $t_j \in [-1, 1]$

- **True functional PCs:**

- Variable 1: $v_{11}(t) = \frac{t + \sin(\pi t)}{s_1}, \quad v_{12}(t) = \frac{\cos(3\pi t)}{s_2}$

- Variable 2:

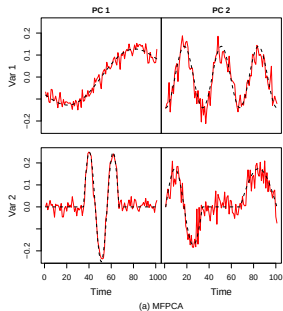
$$v_{21}(t) = \begin{cases} \frac{\sin(3\pi t)}{s_3}, & t \in (-\frac{1}{3}, \frac{1}{3}), \\ 0, & \text{otherwise,} \end{cases} \quad v_{22}(t) = \begin{cases} \frac{\sin(2\pi t)}{s_4}, & t \leq -\frac{1}{3}, \\ \frac{\sin(\pi t)}{s_4}, & t \geq \frac{1}{3}, \\ 0, & \text{otherwise.} \end{cases}$$

Here, s_1, s_2, s_3, s_4 are normalizing constants ensuring unit L^2 norm.

Simulation: Estimation Performance

Scenarios tested:

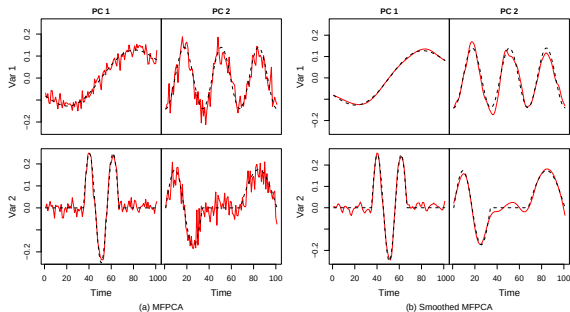
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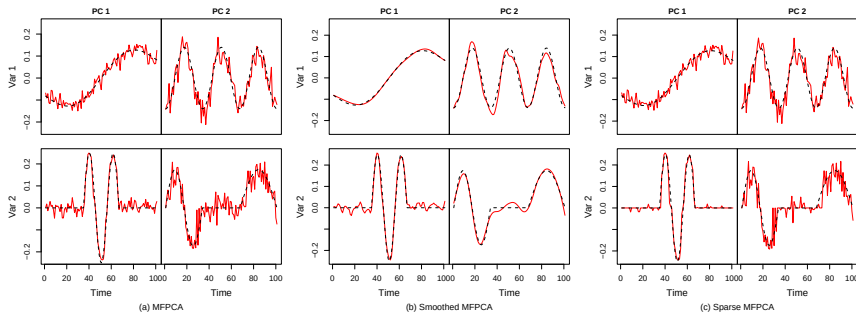
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Simulation: Estimation Performance

Scenarios tested:

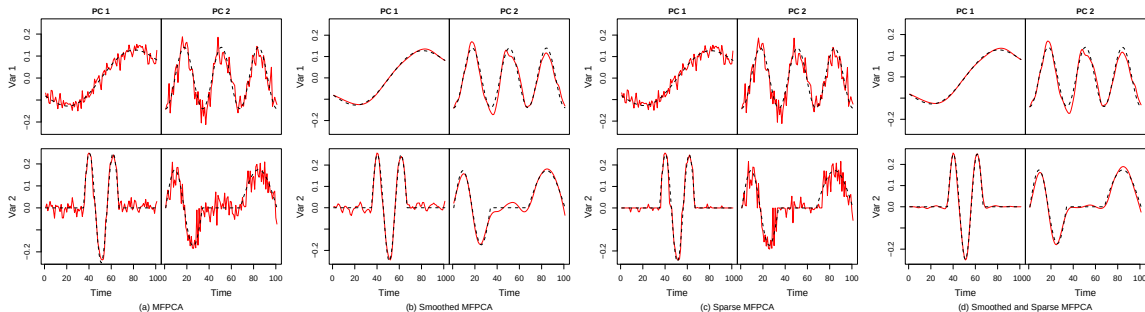
1. Unpenalized Multivariate SVD (baseline)
2. Smoothed Multivariate SVD (smoothness penalty)
3. Sparse Multivariate SVD (sparsity penalty)
4. Sparse + Smoothed Multivariate SVD (combined regularization)



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Accuracy measures:

- ① Variable-wise MSE:

$$\text{MSE}_{k\ell} = \frac{1}{m} \sum_{j=1}^m (\hat{v}_{k\ell}(t_j) - v_{k\ell}(t_j))^2$$

- ② Replication-averaged MSE:

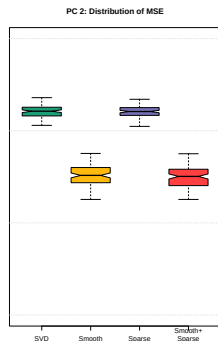
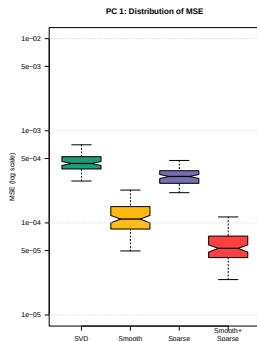
$$\overline{\text{MSE}}_{k\ell} = \frac{1}{R} \sum_{r=1}^R \text{MSE}_{k\ell}^{(r)}$$

- ③ Multivariate MSE:

$$\text{MSE}_{\ell}^{(\text{multi})} = \frac{1}{m} \sum_{j=1}^m \sum_{k=1}^2 (\hat{v}_{k\ell}(t_j) - v_{k\ell}(t_j))^2$$

Simulation: Estimation Performance

- **Performance across four methods (SVD, Smooth, Sparse, Smooth+Sparse):**
 - Smoothness and/or sparsity **reduce MSE** compared to unregularized SVD.
 - **Smooth+Sparse** yields lowest error and most stable estimates.
 - Smooth estimator performs consistently well; sparsity alone less effective (esp. for PC2).
 - Joint regularization achieves best **bias–variance tradeoff**.



PC1: Quartiles and Mean log₁₀(MSE)

Method	Q1	Median	Mean	Q3
SVD	-3.41	-3.35	-3.34	-3.28
Smooth	-4.07	-3.96	-3.92	-3.82
Sparse	-3.57	-3.50	-3.49	-3.43
Smooth+Sparse	-4.38	-4.28	-4.22	-4.14

PC2: Quartiles and Mean log₁₀(MSE)

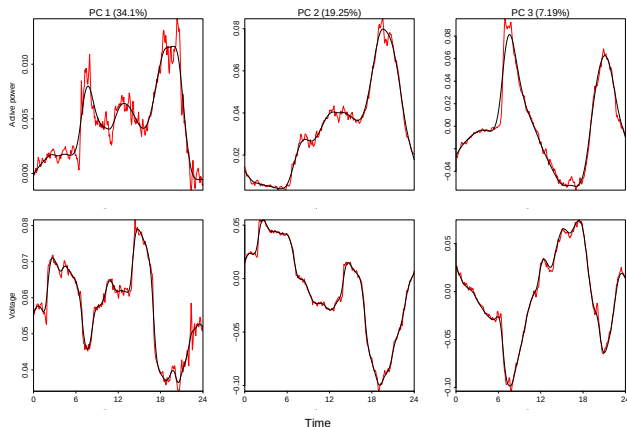
Method	Q1	Median	Mean	Q3
SVD	-2.84	-2.79	-2.79	-2.75
Smooth	-3.56	-3.48	-3.47	-3.40
Sparse	-2.83	-2.79	-2.79	-2.75
Smooth+Sparse	-3.59	-3.50	-3.49	-3.42

Application: Household Power Consumption

- **Dataset:** Bivariate functional data including active power and voltage consumption [Hebrail and Berard, 2012] for one household between December 2006 and November 2010.
- **Scaling:** To equalize the contribution of each variable in the multivariate analysis, we rescale them following [Happ and Greven, 2018].

$$\tilde{X}_j(t_i) = \hat{w}_j^{1/2} X_j(t_i),$$

$$\hat{w}_j = \left(\frac{1}{m} \sum_{i=1}^m \widehat{\text{Var}}(X_j(t_i)) \right)^{-1}.$$



First 3 PCs: MFPCA (red) vs ReMFPCA (black)

Regularization reduces noise while preserving the dominant daily consumption patterns, enhancing interpretability without losing key structure.

Two-way Regularized MFPCA

Conclusion & Future Work

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